

EP3120 - Statistical Mechanics

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Quantum Statistics

Classical versus quantum

- At small scales physical phenomena obey laws of quantum mechanics
 - as observed in hydrogen atom, photoelectric effect, blackbody radiation, etc.
- Quantum mechanics is based on the wave-particle duality, i.e., a particle with momentum p has a De Broglie wavelength

$$\lambda = \frac{h}{p} \quad (1)$$

- This can be thought of as the extent of the wavefunction that describes it's quantum properties
- If the particles are separated by more than the De Broglie wavelength, then their quantum properties are not manifested
- Therefore, classical physics will work if

$$n\lambda^3 \ll 1 \quad (2)$$

- n is the number density of particles

Onset of quantum effects

- Quantum effects will come into the picture when

$$n\lambda^3 \gtrsim 1 \quad (3)$$

- We take the thermal De Broglie wavelength as

$$\lambda = \sqrt{\frac{2\pi\hbar^2}{mk_B T}} \quad (4)$$

- Quantum effects becomes important at the temperature T_0 where

$$n\left(\frac{2\pi\hbar^2}{mk_B T_0}\right)^{3/2} \sim 1 \quad (5)$$

- This is called the quantum degeneracy temperature T_0

Quantum regime

- The degeneracy temperature is

$$k_B T_0 = \frac{2\pi\hbar^2}{m} n^{2/3} \quad (6)$$

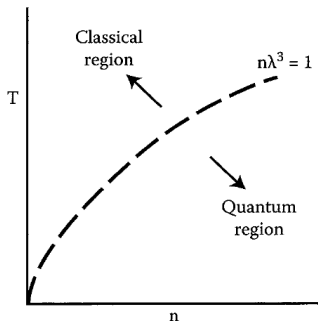


TABLE 14.1 Quantum Degeneracy Temperatures

System	Density (cm^3)	T_0 (K)
H ₂ gas	2×10^{19}	5×10^{-2}
Liquid ⁴ He	2×10^{22}	2
Electrons in metal	10^{22}	10^4

Single particle wave function

- In quantum mechanics a particle is described by a wave function and various properties are obtained by operators acting on that wavefunction
- A position space wavefunction $\psi(\mathbf{r})$ tells us the probability of finding a particle in a particular region of space $\mathbf{r} \rightarrow \mathbf{r} + d\mathbf{r}$, i.e.

$$|\psi(\mathbf{r})|^2 d\mathbf{r} \quad (7)$$

- The normalization condition is

$$\int d\mathbf{r} |\psi(\mathbf{r})|^2 = 1 \quad (8)$$

- The Hamiltonian operator ($\hat{H}$) tells us about the energy of a particle
- The average energy of a particle will be

$$\langle E \rangle = \int d\mathbf{r} \psi^* (\hat{H} \psi) \quad (9)$$

- This is an ensemble average

Two-particle wave function

- Now consider a system having two particles
- To describe this system we need a two-particle wave-function $\psi(\mathbf{r}_1, \mathbf{r}_2)$
- This tells us the probability of finding a particle in the region $\mathbf{r}_1 \rightarrow \mathbf{r}_1 + d\mathbf{r}_1$ and another particle in the region $\mathbf{r}_2 \rightarrow \mathbf{r}_2 + d\mathbf{r}_2$ by

$$|\psi(\mathbf{r}_1, \mathbf{r}_2)|^2 d\mathbf{r}_1 d\mathbf{r}_2 \quad (10)$$

- And the normalization condition becomes

$$\int d\mathbf{r}_1 \int d\mathbf{r}_2 |\psi(\mathbf{r}_1, \mathbf{r}_2)|^2 = 1 \quad (11)$$

- Here the first co-ordinate $\mathbf{r}_1$ can be identified as the position of the “first” particle and the second co-ordinate $\mathbf{r}_2$ can be identified as coordinate of “second” particle
- Of course, each co-ordinate $\mathbf{r}_{1,2}$ can be a 1D, 2D, or 3D position vector

Parity operator

- Define the parity operator $\hat{P}$ as

$$\hat{P}\psi(\mathbf{r}_1, \mathbf{r}_2) = \psi(\mathbf{r}_2, \mathbf{r}_1) \quad (12)$$

- For example, if the two-particle wavefunction were to be a product of two functions $\psi(x_1, x_2) = f(x_1)g(x_2)$ then

$$\hat{P}\psi(x_1, x_2) = f(x_2)g(x_1) \quad (13)$$

- Now clearly $\hat{P}^2 = \hat{I}$, the identity operator
- If an operator acts on a wave-function and returns the same wave-function with a multiplicative factor, i.e.

$$\hat{H}\psi = E\psi \quad (14)$$

- then ψ is an eigenstate of that operator ($\hat{H}$) and the multiplicative constant (E) is the eigenvalue of that eigenstate

Parity operator

- Suppose a two-particle wavefunction $\phi(\mathbf{r}_1, \mathbf{r}_2)$ is an eigenstate of the parity operator with eigenvalue λ

$$\hat{P}\phi = \lambda\phi \quad (15)$$

$$\hat{P}^2\phi = \hat{P}(\lambda\phi) = \lambda^2\phi \quad (16)$$

- But since $\hat{P}^2$ is the identity operator, it means that $\lambda^2 = 1$ and so the eigenvalues of the parity operator are

$$\lambda = \pm 1 \quad (17)$$

- Now let $\rho(\mathbf{r}_1, \mathbf{r}_2)$ be a two-particle wavefunction for identical particles that is an eigenfunction of the Hamiltonian with eigenvalue E , i.e.

$$\hat{H}\rho = E\rho \quad (18)$$

Parity and Hamiltonian

- Suppose we get a wavefunction ν by making $\hat{P}$ act on ρ

$$\nu = \hat{P}\rho \quad (19)$$

- We have changed the parity of ρ by converting it into ν . But it should still remain an eigenfunction of $\hat{H}$ with the same eigenvalue because interchanging the particles positions should not change the energy since the particles are identical

$$\hat{H}\nu = E\nu \Rightarrow \hat{H}\hat{P}\rho = E\nu \quad (20)$$

- Now consider

$$\hat{P}(\hat{H}\rho) = E\hat{P}\rho = E\nu \quad (21)$$

- This can be proved generally for any wavefunction and it implies that

$$\hat{H}\hat{P} = \hat{P}\hat{H} \quad (22)$$

- This is expressed as the parity operator commutes with the Hamiltonian

Parity and Hamiltonian

- Now suppose that ρ is a non-degenerate eigenstate of $\hat{H}$, i.e., there is no other eigenstate with the same eigenvalue E
- Consider

$$\hat{P}\hat{H}\rho = E\hat{P}\rho \quad (23)$$

- Since parity and Hamiltonian commute, so this can also be written as

$$\hat{H}(\hat{P}\rho) = E(\hat{P}\rho) \quad (24)$$

- Since E is a non-degenerate eigenvalue of the Hamiltonian with eigenstate ρ , this means that

$$\hat{P}\rho = \lambda\rho \quad (25)$$

- This shows that a non-degenerate eigenfunction of the Hamiltonian is also an eigenfunction of the parity operator
- We also know that $\lambda = \pm 1$

Fermionic and bosonic states

- The two different eigenvalues of the parity operator correspond to two different types of particles
- Particles have an intrinsic quantum property called spin (with dimensions of angular momentum)
- This spin is quantized in jumps of $\hbar$
- Particles with half integer spins $s = \hbar/2, 3\hbar/2, \dots$ are called fermions
- Particles with integer spins $s = 0, \hbar, 2\hbar, \dots$ are called bosons
- From advanced quantum field theory, Pauli showed that energy eigenstates of fermions are eigenstates of the parity operator with eigenvalue -1
- Similarly, he showed that energy eigenstates of bosons are eigenstates of the parity operator with eigenvalue $+1$
- If there are two single particle wavefunction $f(x)$ and $g(x)$ then the two-particle wavefunction will be

$$\psi(x_1, x_2) = f(x_1)g(x_2) \pm f(x_2)g(x_1) \quad (26)$$

- The plus sign will be for bosons and minus sign will be for fermions

N-particle wavefunction

- Consider an energy eigenfunction of a single particle $u_\alpha(\mathbf{r})$. This is the wave-function of a single particle in the system with a fixed energy
- The α here corresponds to the particular set of quantum numbers that specify this wavefunction
- For ex., for a particle in a 3D box the energy eigenfunction is

$$\psi(\mathbf{r}) = \sqrt{\frac{8}{V}} \sin\left(\frac{n_x \pi x}{L_x}\right) \sin\left(\frac{n_y \pi y}{L_y}\right) \sin\left(\frac{n_z \pi z}{L_z}\right) \quad (27)$$

- with energy

$$E = \frac{h^2}{8m} \left(\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2} + \frac{n_z^2}{L_z^2} \right) \quad (28)$$

- in this case the label $\alpha \rightarrow (n_x, n_y, n_z)$ corresponds to a unique combination of the quantum numbers (n_x, n_y, n_z)

N-particle wavefunction

- Another example is hydrogen atom wavefunctions which are characterized by 4 quantum numbers (n, l, m_l, m_s)
- Principal quantum number $n = 1, 2, 3, \dots$
- Orbital quantum number $l = 0, 1, \dots, n - 1$ (labelled as s, p, ..)
- Magnetic quantum number $m_l = -l, \dots, +l$
- Spin quantum number $m_s = \pm 1/2$
- Electrons are filled accordingly, ex., $3d^9$
- For this system, a single particle wavefunction would be characterized as $\alpha \rightarrow (n, l, m_l, m_s)$
- Assume that these single-particle eigenfunctions are orthonormal

$$\int d\mathbf{r} u_{\alpha}^*(\mathbf{r}) u_{\beta}(\mathbf{r}) = \delta_{\alpha\beta} \quad (29)$$

- Exercise: Construct a 3-particle normalized wavefunction of bosons and fermions from the single-particle wavefunctions u_1 , u_2 , and u_3

Three-particle wavefunction

- For Bosons

$$\psi(\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3) = C \{ u_1(\mathbf{r}_1) u_2(\mathbf{r}_2) u_3(\mathbf{r}_3) + u_2(\mathbf{r}_1) u_1(\mathbf{r}_2) u_3(\mathbf{r}_3) \quad (30)$$

$$+ u_2(\mathbf{r}_1) u_3(\mathbf{r}_2) u_1(\mathbf{r}_3) + u_3(\mathbf{r}_1) u_2(\mathbf{r}_2) u_1(\mathbf{r}_3) \quad (31)$$

$$+ u_1(\mathbf{r}_1) u_3(\mathbf{r}_2) u_2(\mathbf{r}_3) + u_3(\mathbf{r}_1) u_1(\mathbf{r}_2) u_2(\mathbf{r}_3) \} \quad (32)$$

- The normalization constant would be $C = 1/\sqrt{6}$ (assuming all quantum numbers are distinct)
- If two of them are same then $C = 1/\sqrt{6 \times 2}$, if all three are same $C = 1/\sqrt{6 \times 6}$

Three-particle wavefunction

- For fermions

$$\psi(\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3) = C \{ u_1(\mathbf{r}_1)u_2(\mathbf{r}_2)u_3(\mathbf{r}_3) - u_2(\mathbf{r}_1)u_1(\mathbf{r}_2)u_3(\mathbf{r}_3) \quad (33)$$

$$+ u_2(\mathbf{r}_1)u_3(\mathbf{r}_2)u_1(\mathbf{r}_3) - u_3(\mathbf{r}_1)u_2(\mathbf{r}_2)u_1(\mathbf{r}_3) \quad (34)$$

$$- u_1(\mathbf{r}_1)u_3(\mathbf{r}_2)u_2(\mathbf{r}_3) + u_3(\mathbf{r}_1)u_1(\mathbf{r}_2)u_2(\mathbf{r}_3) \} \quad (35)$$

- The normalization constant is $C = 1/\sqrt{6}$
- If any two single-particle functions are the same (ex. $u_1 = u_2$), then the wavefunction will become zero
- This means that only one fermion can occupy a particular quantum state
- This is the Pauli exclusion principle

N-particle wavefunction

- This can be generalized to N particles

$$\psi(\mathbf{r}_1, \dots, \mathbf{r}_N) = \frac{1}{\sqrt{N! \prod_r n_r!}} \sum_P \delta_P P[u_1(\mathbf{r}_1)u_2(\mathbf{r}_2)\dots u_N(\mathbf{r}_N)] \quad (36)$$

- The summation over P represents sum over all permutations of the quantum states $\alpha = 1, 2, \dots, N$ - there are total $N!$ terms
- The factor δ_P is unity for bosonic states and for fermions it is

$$\delta_P = \begin{cases} +1, & \text{for even permutations} \\ -1, & \text{for odd permutations} \end{cases} \quad (37)$$

- The product over r represents distinct quantum states (quantum numbers) and n_r is the number of single-particle wavefunctions in that state
- For ex. if there is a group of 4 single-particle functions with same quantum numbers and another group of 2 single-particle functions with same quantum numbers, then $\prod_r n_r! = 4! \times 2!$
- For fermions, n_r can only be unity

Slater determinant

- For fermions, the wavefunction can be written in the form of a determinant

$$\psi(\mathbf{r}_1, \dots, \mathbf{r}_N) = \frac{1}{N!} \begin{vmatrix} u_1(\mathbf{r}_1) & u_1(\mathbf{r}_2) & \cdot & \cdot & \dots & u_1(\mathbf{r}_N) \\ u_2(\mathbf{r}_1) & u_2(\mathbf{r}_2) & \cdot & \cdot & \dots & u_2(\mathbf{r}_N) \\ \cdot & \cdot & \cdot & \cdot & \dots & \cdot \\ \cdot & \cdot & \cdot & \cdot & \dots & \cdot \\ u_N(\mathbf{r}_1) & u_N(\mathbf{r}_2) & \cdot & \cdot & \dots & u_N(\mathbf{r}_N) \end{vmatrix} \quad (38)$$

- If any two quantum states are same, then those two rows will be identical and the determinant will be zero
- So it will satisfy the Pauli exclusion principle
- This is called as the Slater determinant

Occupation numbers

- A single particle state is labelled by a set of quantum numbers α
- Because of the indistinguishability of quantum particles, the quantum microstate of a system is specified by a unique wavefunction
- That wavefunction is set by the number of particles in each distinct eigenstate α
- That is the set of occupation numbers n_α

$$\{n_1, n_2, \dots\}; \quad \text{where } \alpha = 1, 2, \dots \quad (39)$$

- All the other permutations among the particles are taken care of in the N -particle wavefunction
- If a system has N particles, then

$$\sum_{\alpha} n_{\alpha} = N \quad (40)$$

- If the system has a total energy E and each single-particle state α has energy E_{α} then

$$\sum_{\alpha} n_{\alpha} E_{\alpha} = E \quad (41)$$

Fermionic statistics

- Consider a system composed of fermions. In this case

$$n_{\alpha} = 0 \text{ or } 1 \quad (42)$$

- If we consider a microcanonical ensemble of this system, we will need to calculate the number of microstates Ω such that

$$\sum_{\alpha} n_{\alpha} = N \quad \text{and} \quad \sum_{\alpha} n_{\alpha} E_{\alpha} = E \quad (43)$$

- Counting this is practically impossible
- If we consider a canonical ensemble of this system, then we need to calculate the canonical partition function

$$Q = \sum_{\substack{\{n_{\alpha}\} \\ \sum_{\alpha} n_{\alpha} = N}} e^{-\beta E\{n_{\alpha}\}} = \sum_{\substack{\{n_{\alpha}\} \\ \sum_{\alpha} n_{\alpha} = N}} e^{-\beta(\sum_{\alpha} n_{\alpha} E_{\alpha})} \quad (44)$$

Fermionic statistics

- This summation is practically impossible because of the constraint that $\sum_{\alpha} n_{\alpha} = N$
- The single particle partition function is not of much help because the presence of other particles affects the allowed microstates for one particle
- Instead if we consider the grand canonical ensemble where there is no constraint on the number of particles or on the total energy
- The grand partition function becomes

$$\mathcal{Z} = \sum_{\text{microstates}} e^{-\alpha N - \beta E} \quad (45)$$

$$= \sum_{n_1} \sum_{n_2} \dots \exp[-\alpha \sum_k n_k] \exp[-\beta \sum_k n_k E_k] \quad (46)$$

- The summations over k are over different single-particle quantum states

Fermionic statistics

- This becomes

$$\mathcal{Z} = \sum_{n_1} \sum_{n_2} \dots \left[\prod_k e^{-\alpha n_k - \beta n_k E_k} \right] \quad (47)$$

$$= \prod_k \left[\sum_{n_k} e^{-(\alpha + \beta E_k) n_k} \right] \quad (48)$$

- For fermions $n_k = 0, 1$, so the summation becomes

$$\mathcal{Z} = \prod_k [1 + e^{-\alpha - \beta E_k}] \quad (49)$$

$$= \prod_k [1 + z e^{-\beta E_k}] \quad (50)$$

$$(51)$$

Average occupation number

- The average occupation number of the k^{th} quantum state is

$$\langle n_k \rangle = \frac{1}{\mathcal{Z}} \sum_{\text{microstates}} n_k e^{-\alpha N - \beta E} \quad (52)$$

$$= \frac{1}{\mathcal{Z}} \sum_{\text{microstates}} n_k z^N \prod_j e^{-\beta n_j E_j} \quad (53)$$

$$= \frac{-1}{\beta \mathcal{Z}} \sum_{\text{microstates}} \frac{\partial}{\partial E_k} \left[z^N \prod_j e^{-\beta n_j E_j} \right] \quad (54)$$

$$= -\frac{1}{\beta} \frac{\partial \ln \mathcal{Z}}{\partial E_k} \quad (55)$$

$$= \frac{1}{z^{-1} e^{\beta E_k} + 1} \quad (56)$$

$$= \frac{1}{1 + e^{\beta(E_k - \mu)}} \quad (57)$$

- Exercise: Calculate the average occupation number for bosons

Bosonic statistics

- For bosons, the occupation number $n_k = 0, 1, 2, 3, \dots$ so the bosonic grand partition function becomes

$$\mathcal{Z} = \prod_k \sum_{n_k=0}^{\infty} e^{-(\alpha + \beta E_k)n_k} \quad (58)$$

$$= \prod_k \frac{1}{1 - e^{-\alpha - \beta E_k}} \quad (59)$$

- The average occupation number of the k^{th} quantum state

$$\langle n_k \rangle = -\frac{1}{\beta} \frac{\partial \ln \mathcal{Z}}{\partial E_k} \quad (60)$$

$$= -\frac{1}{\beta \mathcal{Z}} \left(\prod_{m \neq k} \frac{1}{1 - e^{-\alpha - \beta E_m}} \right) \frac{\partial}{\partial E_k} \left(\frac{1}{1 - e^{-\alpha - \beta E_k}} \right) \quad (61)$$

$$= \frac{1}{e^{\beta(E_k - \mu)} - 1} \quad (62)$$

Energy distribution

- The average occupation numbers of fermions and bosons for the k^{th} quantum level can be written as

$$\langle n_k \rangle = \frac{1}{e^{\beta(E_k - \mu)} \pm 1} \quad (63)$$

- where the $+$ sign is for fermions and $-$ sign is for bosons
- If we have to convert this into average occupation number for the i^{th} energy level ϵ_i , then

$$\langle n_i(\epsilon_i) \rangle = \frac{g_i}{e^{\beta(\epsilon_i - \mu)} \pm 1} \quad (64)$$

- where g_i is the degeneracy of the i^{th} energy level